

RH Equivalences: Positivity, Spectral Theory, and the Isometric Section Analogy

Arithmetic Equivalences

The Mertens function $M(x) = \sum_{n \leq x} \mu(n)$ satisfies $M(x) = O(x^{1/2+\varepsilon})$ for all $\varepsilon > 0$ if and only if RH holds. Robin's inequality $\sigma(n) < e^\gamma n \log \log n$ for all $n > 5040$, Lagarias's harmonic number reformulation, and Nicolas's criterion on Euler's totient at primorial values are each equivalent to RH. The Franel–Landau theorem identifies RH with the condition that the discrepancy of Farey fractions from uniform spacing on $[0, 1]$ is $O(n^{1/2+\varepsilon})$, and the Redheffer matrix determinant equals $M(n)$, giving a linear-algebraic restatement.[1]

Analytic Equivalences

The Riesz criterion, Hardy–Littlewood criterion, Speiser's result that $\zeta'(s)$ has no zeros in $0 < \operatorname{Re}(s) < 1/2$ iff RH holds, and Li's criterion (positivity of all $\lambda_n = \sum_\rho [1 - (1 - 1/\rho)^n]$ for $n \geq 1$) are analytic equivalences. The de Bruijn–Newman constant $\Lambda \leq 0$ is equivalent to RH, with Rodgers–Tao proving $\Lambda \geq 0$, so RH is exactly $\Lambda = 0$. [2][1]

The Nyman–Beurling criterion states that RH is equivalent to the characteristic function $\mathbf{1}_{(0,1)}$ lying in the $L^2(0, 1)$ closure of the span of the dilates $\{\rho(\theta/n) : n \in \mathbb{N}, \theta \in (0, 1]\}$, where ρ denotes fractional part. The Baez-Duarte strengthening replaces the full span by the subspace generated by $\{\rho(1/n) : n \in \mathbb{N}\}$ alone. In probabilistic extensions, this closure problem can be reformulated with random dilation factors; equivalences with RH persist for two main classes (exponential and concentrated-around-1 distributions). [3][4][5]

Weil's criterion (1952, extended 1972) is the following precise statement: RH for all Dirichlet L-functions is equivalent to the negativity of the distribution

$$W(g) = \hat{g}(0) + \hat{g}(1) - \sum_{\rho} \hat{g}(\rho)$$

as a functional on the convolution algebra $C_c^\infty(\mathbb{R}_+^\times)$, restricted to functions vanishing on all zeta zeros. Equivalently, Bombieri showed that the associated quadratic functional—obtained from the explicit formula—is positive semidefinite if and only if RH is true, and positive definite on functions supported in a sufficiently small interval regardless. If RH fails with exactly k zeros off the critical line, the number of negative eigenvalues of finite truncations of this quadratic form equals $k/2$. [6][7][8][9]

Spectral and Operator-Theoretic Equivalences

The Hilbert–Pólya conjecture posits a self-adjoint operator whose eigenvalues are the imaginary parts of the nontrivial zeta zeros; no such operator has been explicitly constructed [10]. Berry and Keating proposed that the classical Hamiltonian $H = xp$ on the half-line, with a phase-space cutoff $|x| \geq \ell_x$, $|p| \geq \ell_p$, yields a semiclassical spectrum reproducing the average density of Riemann zeros [11][12]. At the quantum level, normal-order quantization of xp does not exhibit the exact Riemann zeros in its spectrum; realizing them requires additional boundary interaction terms or non-local operators, and the model is not unique [11]. In a refined version, adding interaction terms depending on two boundary wave functions gives a continuum spectrum with discrete embedded bound states whose resonances converge to the Riemann zeros, exploiting a duality symmetry related to the functional equation [13][14].

Montgomery's pair correlation conjecture connects the statistics of spacings between imaginary parts of zeros to the GUE (Gaussian Unitary Ensemble) correlation functions of random matrix theory, a structural similarity that has been verified to high precision numerically but is unproven. The Selberg zeta function associated to a compact hyperbolic surface satisfies the RH analogue

unconditionally, with zeros at $s = 1/2 + it_j$ where $1/4 + t_j^2$ are Laplacian eigenvalues on the surface.[15][1]

Connes's Trace Formula Approach

The most developed spectral program is Connes's formulation on the adèle class space $X_K = \mathbb{A}_K/K^\times$. The spectral realization (Connes 1999) produces the critical zeros as an absorption spectrum: the Weil explicit formula takes the form of a Lefschetz trace formula for the action of the idèle class group C_K on the Hilbert space $L^2_\delta(\mathbb{A}_K/K^\times)_0$. In this Hilbert space setting, only critical zeros appear in the spectrum; noncritical zeros appear as resonances.[16][10]

The cohomological reformulation (Connes–Consani–Marcolli) passes to the cyclic homology $HC_0(D(\mathcal{A}, \varphi))$ of the cokernel of the "restriction map" in the abelian category of cyclic modules. In this setting all zeros contribute to the trace formula, and **RH is equivalent to the positivity of the trace pairing**[17][16]

$$\langle f, f^\sharp \rangle_{H^1} \geq 0 \quad \forall f \in \mathcal{S}(C_K),$$

where $f^\sharp = \Delta^{-1} f^*$ is the involution. This mirrors the two-step structure of the Weil proof for function fields: (1) explicit formula as trace formula, (2) positivity $\text{Tr}(Z \star Z') > 0$ for nontrivial correspondences. In the function field case this positivity is proved via the Riemann–Roch theorem making correspondences effective; the analogue for number fields—identifying the correct class of "principal correspondences"—remains open.[16][17]

The Connes–Consani "Riemann–Roch strategy" (AIM 2018 slides) reformulates RH as the existence of a Riemann–Roch formula $\dim H^0 - \dim H^1 + \dim H^2 = \frac{1}{2} D \cdot D$ on the square of the scaling space, with negativity of the intersection equivalent to RH. The definition of H^1 in this framework is currently an open problem.[18]

Geometric and Cohomological Approaches

Weil proved RH for curves over finite fields using intersection theory on the surface $C \times C$. For a curve $C/\mathbb{F}_q$ of genus g , the key step is

showing that for any nontrivial correspondence $Z \in P(C)$ (divisors modulo trivial and principal ones),[19][20]

$$\mathrm{Tr}(Z \star Z') = d(Z) + d'(Z) - Z \cdot \Delta > 0.$$

This is established by using the Riemann–Roch theorem to make Z effective (adjust degree by trivial correspondences), then applying the Castelnuovo–Severi inequality $\mathrm{def}(D) := 2d_1d_2 - D^2 \geq 0$ for divisors D on $C_1 \times C_2$ [20][21]. The positivity forces all eigenvalues λ_j of Frobenius on $H_{\acute{e}t}^1(\bar{C}, \mathbb{Q}_\ell)$ to satisfy $|\lambda_j| = \sqrt{q}$, which is the Riemann hypothesis[15]. The quadratic form $\mathrm{Tr}(Z_{n,m} \star Z'_{n,m}) = 2gm^2 + 2(1 + q - N)mn + 2gqn^2$ being nonnegative for all integer pairs (n, m) is the discriminant condition $(1 + q - N)^2 \leq 4g^2q$, i.e., $|N - q - 1| \leq 2g\sqrt{q}$ [16].

Deninger has proposed a conjectural cohomology theory for arithmetic schemes X over $\mathrm{Spec}(\mathbb{Z})$, modeled on foliation cohomology of a dynamical system whose "closed orbits" correspond to primes. In this framework, a Lefschetz trace formula of the form[22][23]

$$\zeta(X, s) = \frac{\det_\infty(\frac{s}{2\pi} - \Theta|H^1)}{\det_\infty(\frac{s}{2\pi} - \Theta|H^0) \cdot \det_\infty(\frac{s}{2\pi} - \Theta|H^2)}$$

would hold, where Θ is the generator of the $\mathbb{R}$ -action and $\det_\infty$ is a zeta-regularized determinant[22][23]. RH would then follow from eigenvalues of Θ on H^1 having real part $1/2$, analogous to $|\lambda_j| = \sqrt{q}$ in the function field case. This cohomology is conjectural; the "standard conjectures" analogue needed to run the positivity argument unconditionally remain unproved[24].

The Isometric Section Problem: Structural Location and Precision of Analogy

The proposed connection runs through the spectral/operator-theoretic line. The relevant structural parallel is between two instances of the same abstract problem: **a positivity condition on a quadratic form whose failure obstructs global construction.**

In the isometric section setting, the Killing form B on $\mathfrak{sl}(n, \mathbb{C})$ is indefinite; the Cartan decomposition $\mathfrak{sl}(n, \mathbb{C}) = \mathfrak{su}(n) \oplus \mathfrak{p}$ gives a signature $(n^2 - 1, 0)$ on $\mathfrak{su}(n)$ (negative-definite restricted Killing) and opposite sign on $\mathfrak{p}$. The condition for a right-invariant metric g on $G = SL(n, \mathbb{C})$ to admit an isometric section through the identity—that $g_e + B_k|_{\mathfrak{su}(n)}$ be positive semidefinite—is a quadratic positivity condition on a bilinear form over the Lie algebra. The obstruction to globalizing this from the identity to a smooth section is the integrability condition (bracket-closure / flatness of the associated connection), which generically fails independently of whether the pointwise condition holds.

In Connes's formulation, RH is the positivity $W(f \star f^\sharp) \geq 0$ for all test functions f in the Schwartz space of C_K that vanish on the finite set F of zeros forced by the explicit formula. More precisely, following Connes–Consani, it is the positivity of the trace pairing on H^1 , which is the noncommutative-geometry analogue of the Weil condition $\text{Tr}(Z \star Z') > 0$. The passage from local/finite test function positivity to global positivity on all of $\mathcal{S}(C_K)$ is exactly the point where the proof is incomplete: in the archimedean case (a single place), Connes–Consani (2020) proved a strong inequality showing the Weil local contribution W_∞ exceeds the Sonin trace by a positive term, providing evidence toward Weil positivity but not establishing it in full generality.[7][25][26][27][6][16]

The analogy is therefore precise at the level of **logical shape**: in both cases one has (i) a bilinear/quadratic form on a function space or Lie algebra, (ii) a positivity condition on that form that is equivalent to (or implies) the global object of interest, and (iii) a local-to-global step that constitutes the hard content of the problem. The Hasse–Minkowski theorem for quadratic forms over $\mathbb{Q}$ provides the canonical case where "local positivity everywhere implies global positivity" succeeds; Weil positivity in the number field case is a problem of the same logical type but over a function space, not a finite-dimensional vector space.[28]

What Would Constitute an Actual Equivalence

Promoting this structural analogy to a theorem requires:

1. **An explicit group G :** The natural candidate from the Connes program is a group acting on the adèle class space $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$, specifically the idèle class group $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$ or an extension thereof. This is not a semisimple Lie group of the $SL(n, \mathbb{C})$ type, however; it is a locally compact abelian group. [29][16]
2. **An explicit right-invariant metric g encoding arithmetic data:** The inner product on $L^2_{\delta}(X_{\mathbb{Q}})$ used in Connes's spectral realization is determined by the weight δ , which in turn encodes the archimedean place. The metric would need to be constructed so that positive semidefiniteness of $g_e + B_k|_{\mathfrak{su}(n)}$ is equivalent to a statement about the spectrum of the scaling action[10].
3. **An explicit identification of the section obstruction with a zero off the critical line:** In the correspondence $\mathrm{Tr}(Z \star Z') > 0 \Leftrightarrow |\lambda_j| = \sqrt{q}$, the failure of positivity is pinned to specific Frobenius eigenvalues. The analogous identification would require showing that failure of positive semidefiniteness of the metric restriction corresponds to a specific eigenvalue of the scaling operator lying off the line $\mathrm{Re}(s) = 1/2$.

Without items (1)–(3) made explicit and verified, the parallel is a structural analogy, not a theorem. The gap is not terminological but substantive: "same shape of argument" and "logically equivalent" differ by the entire content of the construction. The Weil proof for curves succeeds precisely because steps (1)–(3) have concrete realizations—the curve C , the Jacobian pairing, and Frobenius eigenvalues—that can be handled by Riemann–Roch. The number field case lacks a proved analogue of Riemann–Roch on the adèle class space.[18][16]

Status of Component Claims

Claim	Status
$M(x) = O(x^{1/2+\varepsilon}) \Leftrightarrow \text{RH}$	Proved equivalent [1]
Robin's / Lagarias's / Nicolas's criteria $\Leftrightarrow \text{RH}$	Proved equivalent [1]
Weil criterion (positivity of quadratic functional $\Leftrightarrow \text{RH}$)	Proved equivalent [8][9]
Nyman–Beurling / Baez-Duarte density $\Leftrightarrow \text{RH}$	Proved equivalent [4][30]
Connes trace formula formulation, $\text{RH} \Leftrightarrow$ positivity of H^1 pairing	Formulated; equivalence established assuming Meyer's analytic framework [16][10]
Connes–Consani Weil positivity at archimedean place	Strong inequality proved (2020); full Weil positivity unproved [25][26][27]
Berry–Keating $H = xp$ model realizing exact zeros	Semiclassical average proven; exact spectrum requires non-unique boundary terms [11][13]
Deninger cohomology giving RH via Lefschetz trace	Conjectural; requires unproved standard conjectures analogue [22][23]
Weil RH for curves (function fields)	Proved [20][21]
Isometric section positivity $\Leftrightarrow \text{RH}$	No explicit construction known; structural analogy only

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